

MARTIN BISSONNETTE

Nanotechnology Engineering Student · University of Waterloo

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⚙️ SKILLS

Lab Experience: Photolithography, Wet/Dry Etching, Cleanroom Protocols, Organic Chemistry, Optical Alignment, Laser Operation & Safety

Lab Equipment: AFM, SEM, Lithography (UV & 2-photon), XRD, PVD, CVD, RIE, Spectroscopy (UV-Vis, FTIR, Raman, Ellipsometer), TGA, DSC

Software & Simulation: SolidWorks, AutoCAD, Fusion, OnShape, FEMM, KLayout, KiCAD, Gwyddion, ImageJ/Fiji

Hardware: Microcontrollers (Arduino, ESP), Raspberry Pi, CNC systems and GCODE based systems

Programming: Python, MATLAB, Excel VBA, C++, HTML, CSS

Languages: Fluent in English and French (DELF Certified)

📁 WORK EXPERIENCE

Packaging & Assembly Engineer

January 2026 – May 2026

Irradiant Technologies

Boston, Massachusetts

- Developed semiconductor packaging methods for proprietary nano-porous scaffold materials, increasing quality, yield and throughput by 10x and enabling the company to acquire a \$500K contract with Coherent.
- Developed novel dicing methods that preserved the optical clarity of porous glass materials, enabling three-dimensional optical paths and reducing optical losses.
- Advanced the development of high-fidelity 3D micro-electronic systems under DARPA's AMME program, supporting proposals that secured over \$4M in contract awards.
- Assisted in the design and validation of an optical measurement system for insertion loss and beam quality of printed photonic waveguides, enabling data-driven optimization of print geometry and process conditions.

Materials & Characterization Engineer

September 2025 – December 2025

Irradiant Technologies

Boston, Massachusetts

- Designed and produced photosensitive silver reduction materials for use in multi-photon 3D lithography systems, optimizing for conductivity, surface quality, and sub-micron resolution.
- Conducted several AFM and SEM measurement cycles at Harvard CNS (Trained User) that directly contributed to a \$1.8M milestone payment and multiple customer developments.
- Implemented a multi-FOV stitching feature for the point-scan lithography system, enabling printing of large-scale conductive structures at nanoscale resolution, allowing the creation of GHz RF structures.
- Co-authored a low-cost open-source two-photon lithography system for academic research – [ResearchGate](#).

Packaging & Process Engineering Intern

January 2025 – May 2026

C2MI

Bromont, Quebec

- Performed microfab and testing in Class 1000 cleanrooms using semiconductor and optical packaging tools.
- Performed R&D on photonic wire bonds as an alternative photonic interconnect method in collaboration with IBM; designed a photonic waveguide chip using KLayout for testing insertion losses of optical interconnects
- Developed a laser-induced graphene (LIG) process for PCB manufacturing, enabling flexible, high-conductivity electroplating on sub-mm traces comparable to bulk electrodes.
- Documented and tested a wafer-level packaging method to embed chips into remolded wafers, enabling further processing and higher package density.
- Optimized wafer-level packaging for delicate materials (InP), enabling backside grinding with increased yield.
- Characterized materials and assembly quality using optical and electro-mechanical analysis techniques.

Part-Time Undergraduate Research Assistant (URA)

May 2025 – August 2025

SMART LAB

Waterloo, Ontario

- Synthesized hydrogels and characterized materials to evaluate properties for targeted drug delivery.

🔬 PROJECTS

UW Hacker Fab — Project Lead

December 2023 – Ongoing

- Leading design and fabrication of a low-cost OPU based AFM that re-purposes salvaged DVD optical pickups as a sub-um deflection sensors for thin-film metrology.
- Currently fabricating a piezo-based AFM scanning stage adapted from a published [design](#).
- Contributed to process design and fabrication of test structures, with UW HackerFab's first MOSCAPs currently underway.

🎓 EDUCATION

University of Waterloo

Bachelor of Applied Science in Honours Nanotechnology Engineering 3.6/4.0 GPA

Fall 2023 – Spring 2028

Ontario, Canada